1.1 Problems — Proofs, logic and sets

Attempt each problem before opening solutions.md. Write every proof in full sentences. $\mathbb{N} = \{1, 2, 3, \dots\}$.

Warm-up

1. (a) Use truth tables to show that $P \Rightarrow Q$, $\neg Q \Rightarrow \neg P$ and $\neg P \vee Q$ are logically equivalent.

(b) Show that $P \Rightarrow Q$ is not equivalent to its converse $Q \Rightarrow P$.
(c) Show that $(P \Rightarrow Q) \wedge (P \Rightarrow R) \equiv P \Rightarrow (Q \wedge R)$.

2. For each statement, decide whether it is true, and write its negation with no $\neg$ in front of a quantifier.

(a) $\forall x \in \mathbb{R}\ \exists y \in \mathbb{R} : y > x$.
(b) $\exists y \in \mathbb{R}\ \forall x \in \mathbb{R} : y > x$.
(c) $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N : \dfrac{1}{n} < \varepsilon$.

3. Let n be an integer. Prove that if n is odd then n^2 is odd, and that if n^2 is even then n is even. Which proof technique does the second part use?

4. Prove, by double inclusion, that for all sets A, B, C:

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$

5. Decide whether each function is injective and whether it is surjective. Prove your answers.

(a) $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = 2n + 1$.
(b) $g : \mathbb{R} \to \mathbb{R}$, $g(x) = x^3 - x$.
(c) $h : \mathbb{R} \to (0, \infty)$, $h(x) = e^x$.

6. Prove by induction that $2^n > n^2$ for every integer $n \geq 5$.

Core

7. Prove that $\sqrt{3}$ is irrational.

8. Prove that there are infinitely many primes of the form $4k + 3$.

9. Prove by strong induction that every integer $n \geq 12$ can be written as $n = 4a + 5b$ with a, b non-negative integers.

10. Let $f : A \to B$.

(a) Prove that $f(f^{-1}(Y)) \subseteq Y$ for all $Y \subseteq B$, with equality for every Y if and only if f is surjective.
(b) Prove that $X \subseteq f^{-1}(f(X))$ for all $X \subseteq A$, with equality for every X if and only if f is injective.

11. Let $f : A \to B$ and $g : B \to C$, and suppose $g \circ f$ is bijective.

(a) Prove that f is injective and g is surjective.
(b) Give an example where neither f nor g is bijective.

12. On $\mathbb{R}$ define $x \sim y$ if $x - y \in \mathbb{Q}$.

(a) Prove that $\sim$ is an equivalence relation.
(b) Prove that every equivalence class is countable.
(c) Deduce that the set of equivalence classes $\mathbb{R}/\sim$ is uncountable.

13. On $X = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ define $(a, b) \sim (c, d)$ if $ad = bc$.

(a) Prove that $\sim$ is an equivalence relation. Take care with transitivity.
(b) Define $[(a, b)] + [(c, d)] := [(ad + bc, bd)]$. Prove that this is well defined.

14. Consider divisibility, $a \mid b$, on $\mathbb{N}$.

(a) Prove that it is a partial order but not a total order.
(b) Does $\mathbb{N}$ have a least element? A greatest element?
(c) Find all minimal and all maximal elements of $\{2, 3, \dots, 10\}$. Does this set have a greatest element?

15. (a) Give an explicit injection $\mathbb{Z} \times \mathbb{Z} \to \mathbb{N}$ and prove that it is injective.

(b) Deduce that $\mathbb{Q}$ is countable.

16. Prove that if $A_1, A_2, \dots$ are countable sets, then $\bigcup_{n=1}^{\infty} A_n$ is countable. Point out where you use the axiom of (countable) choice.

17. Prove Cantor's theorem: for any set A there is no surjection $A \to \mathcal{P}(A)$.

Challenge

18. Prove the Schröder–Bernstein theorem: if there are injections $f : A \to B$ and $g : B \to A$, then there is a bijection $A \to B$.

19. (a) Write down an explicit bijection $[0, 1] \to [0, 1)$.

(b) Using Schröder–Bernstein, prove that $|\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$.

20. Use Zorn's lemma to prove that every vector space has a basis. State precisely which partially ordered set you use and verify every hypothesis of Zorn's lemma.

21. (a) Prove that the set of sequences $(x_n)_{n \geq 1}$ with $x_n \in \{0, 1\}$ that are *eventually zero* is countable.

(b) Prove that the set of all functions $\mathbb{N} \to \mathbb{N}$ is uncountable.

22. Let $n \in \mathbb{N}$ and let S be any set of $n + 1$ numbers chosen from $\{1, 2, \dots, 2n\}$.

(a) Prove that S contains two numbers that are coprime.
(b) Prove that S contains two distinct numbers, one of which divides the other.